

Ex No: 2:**Table Creation and Data Insertion:**

SQL

Create table Student(StudentID int, Name varchar(30), Age int);

Create table Courses (CourseID int, CourseName varchar(20));

Create table Enrollments (EnrollmentID int, StudentID int, CourseID int, Grade varchar(5));

Insert into Student values(1,'Alice', 20);

Insert into Courses values (101,'Database Management');

Insert into Enrollments values(1,1,101,'A');

Inner Join:

```
SELECT Students.StudentID, Students.Name, Students.Age, Courses.CourseID,  
Courses.CourseName, Enrollments.Grade FROM Students INNER JOIN Enrollments ON  
Students.StudentID = Enrollments.StudentID INNER JOIN Courses ON  
Enrollments.CourseID = Courses.CourseID;
```

Output:

	StudentID	Name	Age	CourseID	CourseName	Grade
	1	Alice	20	1	Math	A
	1	Alice	20	2	English	B
	2	Bob	22	1	Math	A-
	3	Charlie	21	3	History	B+
	3	Charlie	21	2	English	A

Left Join:

```
SELECT Students.StudentID, Students.Name, Students.Age, Courses.CourseID,  
Courses.CourseName, Enrollments.Grade FROM Students LEFT JOIN Enrollments ON  
Students.StudentID = Enrollments.StudentID LEFT JOIN Courses ON Enrollments.CourseID  
= Courses.CourseID;
```

Output:

	StudentID	Name	Age	CourseID	CourseName	Grade
	1	Alice	20	1	Math	A
	1	Alice	20	2	English	B
	2	Bob	22	1	Math	A-
	2	Bob	22	NULL	NULL	NULL
	3	Charlie	21	3	History	B+
	3	Charlie	21	2	English	A

Right Join:

```
SELECT Students.StudentID, Students.Name, Students.Age, Courses.CourseID,  
Courses.CourseName, Enrollments.Grade FROM Courses RIGHT JOIN Enrollments ON  
Courses.CourseID = Enrollments.CourseID RIGHT JOIN Students ON  
Enrollments.StudentID = Students.StudentID;
```

Output:

StudentID	Name	Age	CourseID	CourseName	Grade
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1	Alice	20	1	Math	A
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1	Alice	20	2	English	B
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2	Bob	22	1	Math	A-
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3	Charlie	21	3	History	B+
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3	Charlie	21	2	English	A
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